

Analyzing the Impact of Car Features on Price and Profitability

Project description

This project focuses on analyzing a dataset of car models and their specifications to understand how various features impact a car's market price (MSRP) and potential profitability. The aim is to help car manufacturers make data-driven decisions in pricing and product development, aligning offerings with consumer demand and market trends

Approach :

- Start with a clear and concise summary of your insights
- Use clear and concise language
- Use visual aids
- Provide context
- Highlight key findings
- Consider the audience

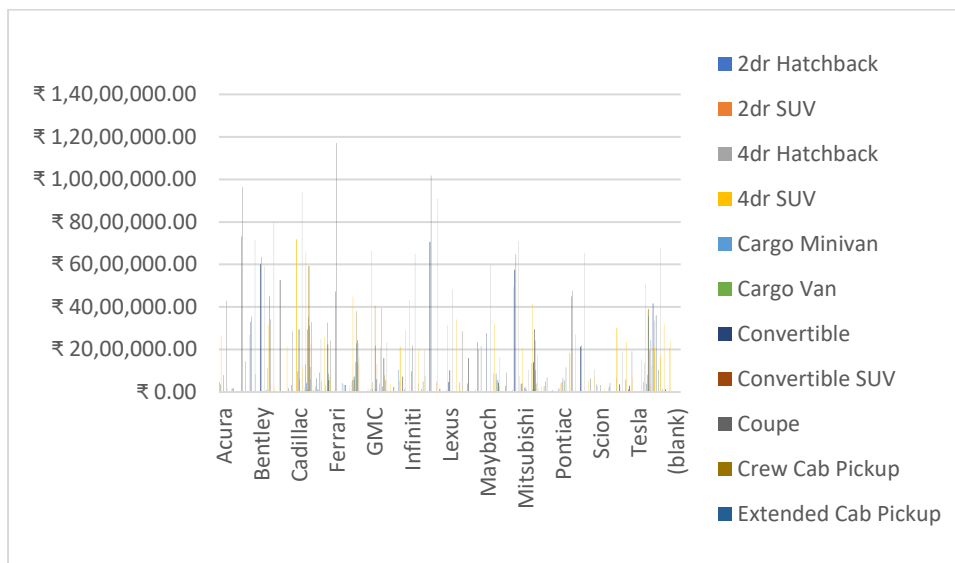
Tech Stack Used :

1. MS Excel

Task 1: Distribution of Car Prices by Brand and Body Style

Insight:

- Luxury brands like **BMW**, **Audi**, and **Mercedes-Benz** tend to have higher total MSRP values across most body styles (especially **Sedan**, **Coupe**, and **Convertible**).
- Economy brands like **Toyota** or **Honda** contribute more significantly to **SUV** and **Hatchback** segments but with lower average price points.
- Brands with a broader model lineup across different styles show a more distributed price spectrum.

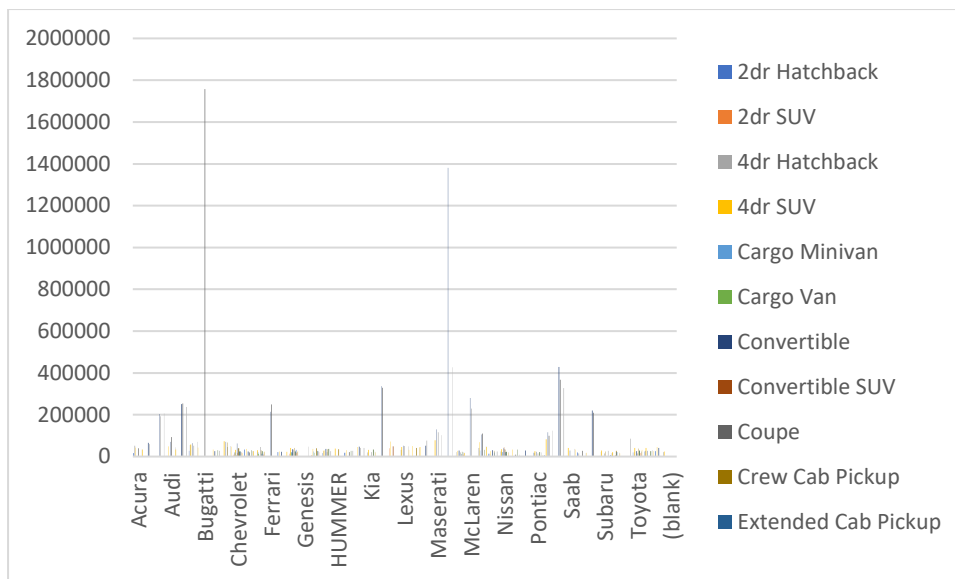


Transmission Type	Engine Fuel Type	Year
AUTOMATED_MANUAL	diesel	1990
AUTOMATIC	electric	1991
DIRECT_DRIVE	flex-fuel (premium un...	1992
MANUAL	flex-fuel (premium un...	1993
UNKNOWN	flex-fuel (unleaded/E85)	1994
(blank)	flex-fuel (unleaded/n...	1995
	natural gas	1996
	premium unleaded (r...	1997

Task 2: Average MSRP by Brand and Body Style

Insight:

- **Luxury brands** (e.g., **Mercedes-Benz**, **Porsche**) have higher average MSRPs, particularly in **Convertible** and **Coupe** segments.
- **Compact and Hatchback styles** from **Ford**, **Hyundai**, or **Kia** show relatively lower average prices.
- **Body style** heavily influences pricing — **Convertibles** and **Coupes** are often more expensive, regardless of brand.

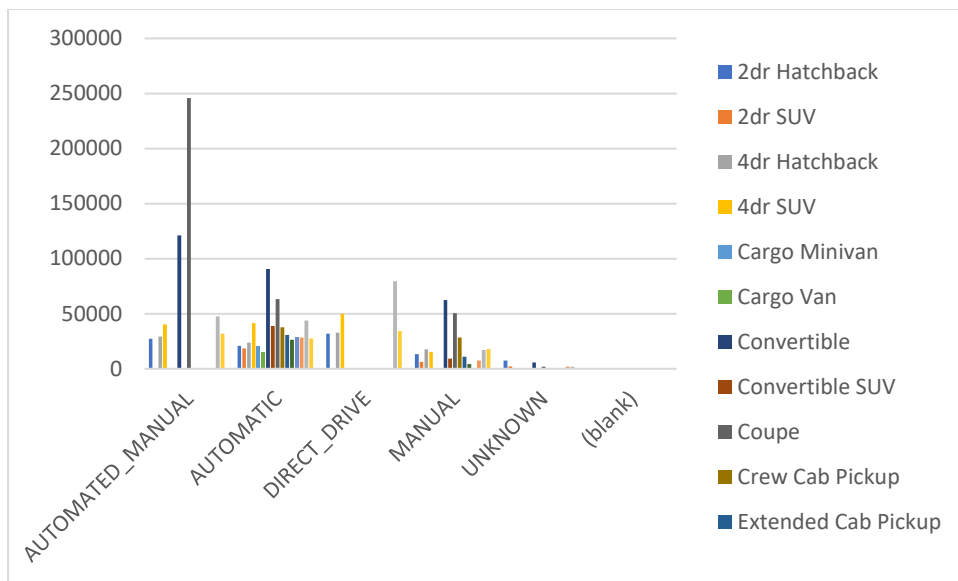


Transmission Type	Year
AUTOMATED_MANUAL	1990
AUTOMATIC	1991
DIRECT_DRIVE	1992
MANUAL	1993
UNKNOWN	1994
(blank)	1995
	1996
	1997

Task 3: Transmission Type vs MSRP by Body Style

Insight:

- **Automatic transmission** vehicles generally have a **higher average MSRP**, especially in **SUVs** and **Luxury Sedans**.
- **Manual transmission** vehicles are cheaper and more commonly found in **Compact cars** and **Performance Coupes**.
- Transmission preference reflects target markets—manuals for driving enthusiasts, automatics for convenience seekers.

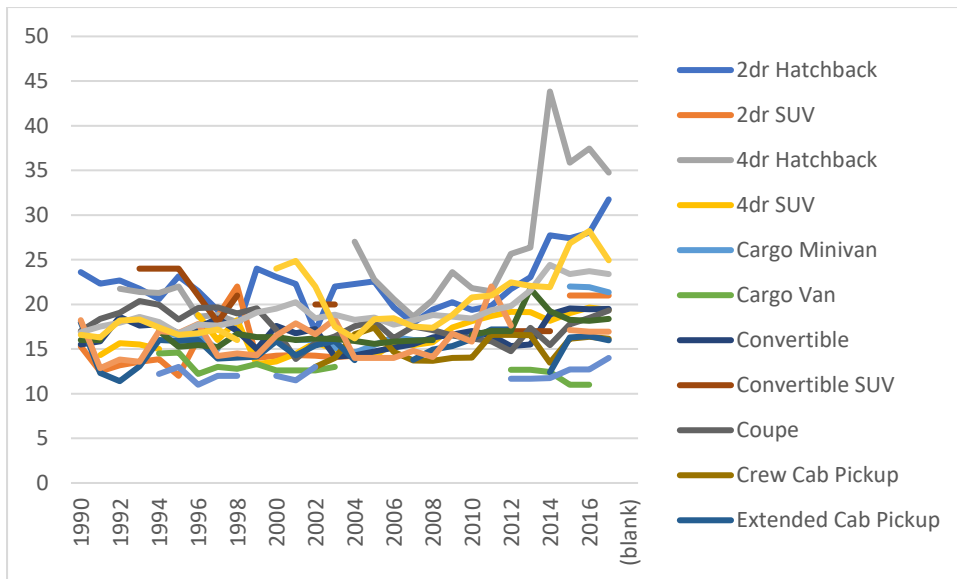


Make	Year
Acura	1990
Alfa Romeo	1991
Aston Martin	1992
Audi	1993
Bentley	1994
BMW	1995
Bugatti	1996
Buick	1997

Task 4: Fuel Efficiency Trends by Body Style and Year

Insight:

- **Fuel efficiency (MPG)** has improved gradually over newer model years, especially in **Sedans** and **Compact SUVs**.
- **Larger vehicles** like **SUVs** and **Trucks** still lag in MPG despite tech improvements.
- The **hybridization trend** likely improves MPG in recent years for all body styles.



Make	Engine Fuel Type
Acura	diesel
Alfa Romeo	electric
Aston Martin	flex-fuel (premium unleaded)
Audi	flex-fuel (premium unleaded)
Bentley	flex-fuel (unleaded/E85)
BMW	flex-fuel (unleaded/natural gas)
Bugatti	natural gas
Buick	premium unleaded (regular)

Task 5: Horsepower, MPG, and Price Variation by Brand

Insight:

- Brands like Tesla, BMW, Audi, and Mercedes-Benz combine high horsepower and higher MSRPs, with moderate to good MPG.
- Japanese brands like Toyota and Honda focus on high fuel efficiency and balanced horsepower, with lower price tags.
- Brands targeting performance (e.g., Dodge, Chevrolet) may have high horsepower but lower MPG.

Car Brand Comparison: Performance and Pricing

